MTH 335(A) Dr. James Ashe

**Homework 6** Due date: Friday $26^{\text{th}}$ April, 2013 **Solutions**

**Definition.** A *group homomorphism* is a function $f : G_1 \to G_2$ between two groups that preserves the group operation; i.e., $f(ab) = \underbrace{f(a)f(b)}_{\text{operation of } G_2}$.

ex. The mapping $f : \mathbb{Z} \to \mathbb{Z}_n$ where $\mathbb{Z}$ and $\mathbb{Z}_n$ are groups under addition and modular addition respectively. Since $f(a+b) = (a+b) \bmod n = a \bmod n + b \bmod n$.

**Definition.** Let $G$ be a group a $H$ a subset of $G$. For any $a \in G$, the set $aH = \{ah | h \in H\}$ is called the *left coset of $H$ in $G$ containing $a$*. $Ha = \{ha | h \in H\}$ is called the *right coset of $H$ in $G$ containing $a$*.

**Lemma.** *For $H$ a subgroup of $G$ and $a, b \in G$, the following properties of cosets hold:*

***1.*** $a \in aH$

*Proof.* $e \in H$ since $H \leq G$. So $a = ae \in aH$. □

***2.*** $aH = H$ *if and only if* $a \in H$.

*Proof.* Let $aH = H$. Then $a = ae \in aH = H$. Suppose that $a \in H$. $aH \subseteq H$ since $H$ is closed, and $H \subseteq aH$ since if $h \in H$ then $a^{-1}h \in H$. Thus $h = eh = \left(aa^{-1}\right)h = a\left(a^{-1}h\right) \in aH$. □

***3.*** $aH = bH$ *or* $aH \cap bH = \emptyset$

*Proof.* Suppose that $aH \cap bH \neq \emptyset$. Then there exists an $x \in aH \cap bH$, and there are $h_1, h_2 \in H$ such that $x = ah_1 = bh_2$. Thus $a = xh_1^{-1} = bh_2h_1^{-1}$. So then $aH = \left(bh_2h_1^{-1}\right)H = bH$ since $h_2, h_1^{-1} \in H$ (because $H \leq G$). □

***4.*** $|aH| = |Ha| = |bH| = |H|$.

*Proof.* Two sets have equal order if and only if there is a bijection between them. Let $h \mapsto ah$ define a map from $H$ to $aH$. $|aH| \leq |H|$ by definition, so the $h \mapsto ah$ is onto. Because $G$ is a group there exists an $a^{-1}$. So then there exists an inverse map $a^{-1}(ah) \mapsto h$ so that map is $1-1$. Thus it is a bijection and $|H| = |aH|$. Similarly a bijection exists for $bH$ and $Ha$, and $|aH| = |Ha| = |bH| = |H|$. □

**Theorem.** ***Lagrange's Theorem:*** $|H| \,|\, |G|$***.*** *If $G$ is a finite group and $H$ is a subgroup of $G$, then $|H|$ divides $|G|$. (also $\dfrac{|G|}{|H|}$ is the number of distinct left cosets in $G$).*

*Proof.* Let $a_1H, a_2H, \ldots, a_nH$ be a list of distinct left cosets of $G$. For each $a \in G$, $aH = a_iH$ for some $1 \leq i \leq n$. By part 1 of the lemma, $a \in aH = a_iH$. Thus each $a \in G$ is in on of the cosets from the list, and we can write

$$G = a_1H \cup a_2H \cup \cdots \cup a_nH.$$

By part 3 of the lemma, $a_iH \cap a_jH = \emptyset$ for each $i$ and $j$ from the list. So then

$$|G| = |a_1H| + |a_2H| + \cdots + |a_nH|.$$

By part 4, $|a_iH| = |H|$. So we have that $|G| = n\,|H|$ which implies that $|H| \,|\, |G|$. □

**1.** Prove that if $|G| \leq 4$ then $G$ is abelian. Then prove that if $|G| < 6$ then $G$ is abelian.

*Proof.* Suppose that $|G| \leq 4$ and that $ab \neq ba$ for some $a, b \in G$. Since $ab \neq ba$ then $a \neq b$ and neither $a$ or $b$ are the identity. So then $e, a, b, ab, ba$ are all distinct elements in $G$ but then $|G| > 4$ which is a contradiction.

Alternatively, since $G$ is nontrivial it has an element $a$ such that $a \neq e$. By Lagrange's Theorem, $|a| = 4$ or 2. If $|a| = 4$ then $G$ is cyclic and thus abelian. If all of its nontrivial elements have order 4 then $e = ab\,(ab) \Rightarrow ba = ba\,(abab) = b\,(aa)\,bab = bbab = ab$. Since $aa = e$ and $bb = e$.

For the second part we need only show $G$ is abelian when $|G| = 5$. 5 is prime so then if $a \in G$ and $a \neq e$ then$|a|\,|5$. So then $G$ is cyclic. □

**2.** Let $f : G_1 \to G_2$ be a group homomorphism and let $H \leq G_2$. Prove that $f^{-1}(H) = \{g \in G_1 | f(g) \in H\}$ is a subgroup of $G_1$.

*Proof.* $e \in f^{-1}(H)$ so $f^{-1}(H) \neq \emptyset$. Also $f^{-1}(H) \subseteq G_1$ by definition. Now let $a, b \in f^{-1}(H)$. By definition of $f^{-1}(H)$, $f(a), f(b) \in H$, and thus $f(b^{-1}) \in H$ since $H$ is a subgroup. So then $f(ab^{-1}) = f(a)f(b)^{-1} \in H$ since $f$ is a homomorphism and $H$ is a subgroup, which implies that $ab^{-1} \in f^{-1}(H)$. □